Linear Fairness: Making Per-Class A Priority

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Abstract

Class bias is a pervasive phenomenon observed across machine learning wherein models learn features that skew their predictions to favor certain classes. The mechanisms that cause these learning imbalances however, remain largely unknown. In this paper we examine per-class fairness specifically through the lens of the Max-Min class gap. We leverage the ideas of Neural Collapse to observe how a convex combination of the input data with points projected onto vertices of an Equiangular Tight Frame (ETF) affects class fairness. We find this convex combination improves similarity of average convergence speed between classes and dive into a preliminary examination of the causes. We discover initialization as a potential cause of class bias in the linear regime. Our study explores both projection and initialization methods for reducing class bias, and sets the stage for future research in the underexplored area of continuous per-class fairness. All code is available at: `github`

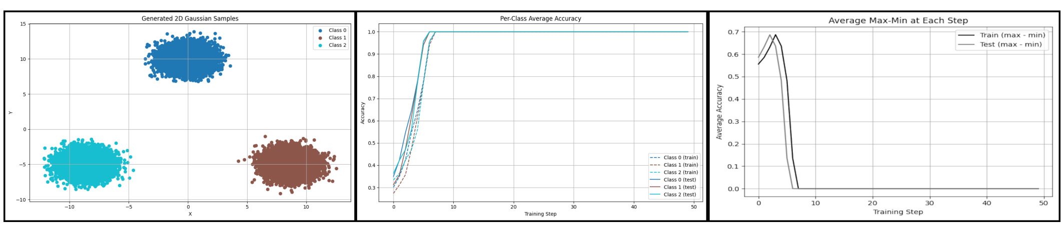

Figure 1: Linear classification of three equidistant 2D data clusters centered about the origin still exhibits unfairness. Although average training speed appears uniform, the Max-Min shows a 68% difference in accuracy between the best and worst classes on average throughout training.

1 Introduction

Class bias in machine learning is where models learn to favor certain groups of inputs over others. As AI use continues to grow it becomes increasingly important that the models we deploy are fair. There is little debate in the field that class bias is an issue [5, 17]. It's intuitive to want your model to learn fairly in respect to all input groups. Since the problem is so well known it raises the question of why the many research efforts in the area have fallen short. Much work has been done on empirical methods to mitigate biased outcomes in larger networks[20, 14]. These valuable studies tend to lack provable results due to the complexity of the networks. This approach can lead to unexplained insights that may not transfer, and improve fairness due to spurious correlation. Most theoretically based work has been done on convergence speed. Many studies focus on data disparities and the convergence dynamics of Neural Networks generally. However, there seems to be a lack of studies accounting for how multiple classes may cause differences in per-class learning speed.

There is an unusual void surrounding the study of continuous per-class fairness specifically. This is a void that must be addressed if we hope to move forward and create provably fair models. Due to the lack of research on continuous per-class fairness our study begins by showing that even in the simplest possible model, class bias arises. Starting in the linear regime retains provability and provides a starting point for theoretically based research in the area. We explore methods to understand and mitigate class bias in this regime. We also gather insights which can later be applied to larger networks. Addressing the problem of class bias in this bottom up approach allows us to build our understanding and future methods on solid foundations with provable results. Our preliminary results show initialization as a likely cause to explain class bias in our experiments. Limitations of this finding and future extensions to this work are outlined in later sections.

2 Background

Many empirical studies on class bias have produced meaningful results[17, 18, 19, 16, 11]. These results are generally limited in their scope and provability due to the intractable nature of large models[15, 12]. Lacking a complete understanding of the networks in which the methods are deployed is a limitation of past research that we address by operating in the linear regime[3, 4, 2].

Many theoretical studies also focus on convergence speed and optimization dynamics [26, 10, 24, 22, 21, 25]. These studies do not explicitly consider class disparities and their implications for fairness across multiple classes during the training process. This is a crucial gap because model bias can emerge directly from the training dynamics, especially when using regularization methods like early stopping, even if the models might appear fair at full convergence [9, 7, 23, 13].

There are also many studies on imbalanced data affecting class fairness [8, 29, 30]. Recent studies also allude to spectra as an indicator for class unfairness even with balanced data [1]. In our study we show that even with balanced spectra class bias emerges. There is currently no proven cause for training unfairness in the multi-class case of per-class bias, even in networks as simple as a linear classifier. Developing an understanding of why unfairness emerges in this fundamental case can lead to a unifying theory of class bias that is provable and developed from first principles, similar to concepts like Neural Collapse [6, 27, 28].

3 Even Simple Models Exhibit Unfairness

3.1 Model

Our network can be viewed as a function $f : \mathbb{R}^2 \to \mathbb{R}^3$ that maps inputs $x \in \mathbb{R}^2$ to outputs $y \in \mathbb{R}^3$ via a weights matrix W^{3x2} as shown below:

$$Wx + b = y$$

$$\begin{bmatrix} w_{0,0} & w_{0,1} \\ w_{1,0} & w_{1,1} \\ w_{2,0} & w_{2,1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$$

Our inputs consist of three sets of $n = 10,000$ points randomly sampled from three 2D Gaussian distributions in a 70:30 train-test split. Our distributions are defined by mean vectors $\mu \in \mathbb{R}^2$ and covariance matrices $\sigma \in \mathbb{R}^{2x2}$. When referencing distributions we use $\{\mu_0, \mu_1, \mu_2\}$ to refer to the means of classes 0-2 respectively. For the experiments shown below the covariance matrices are all the identity matrix. The class labels are one-hot encoded vectors and we call the the true labels y and the model predictions $\hat{y}$. For a correct classification we have:

$$argmax(y) = argmax(\hat{y}).$$

We use Mean Squared Error Loss:

$$L = \frac{1}{n} \sum_{i=1}^{n} (\hat{y_k} - y_k)^2$$

and Stochastic Gradient Descent with learning rate $\eta = 0.01$ for most experiments where:

$$(\eta * \nabla L) + W \rightarrow W.$$

We vary learning rates and optimizers in experiments shown on GitHub to verify bias does not occur due to these factors. Biases are manually initialized to 0.

3.2 Experiments

The below experiments are run on 100 seed initializations, with 50 training steps each. The loss at each training step is averaged over the 100 seeds to show reproducible bias in convergence speed. We also calculate and average the maximum - minimum per-class classification accuracy at each training step. This allows us to observe the difference between the best performing class and the worst performing class throughout training.

We begin with 3 linearly separable classes equidistant from each other and evenly spaced about the origin such that $\|\mu_0\|_2 = \|\mu_1\|_2 = \|\mu_2\|_2$. This configuration was motivated by Neural Collapse, which indicates the last layer of neural networks converge to an Equiangular Tight Frame(ETF). By beginning with data we believe to be "fair" and making alterations we can observe which changes to data cause class imbalance. We find that with equidistant points the average convergence speed appears uniform among the classes but the Max-Min is still considerable(Figure 1). Next we move each class 2 times farther and find that even such small perturbations induce bias, as the farther classes converge much faster on average(Figure 3). We also summarize the results below.

Table 1: Results of doubling the mean of only one class. We show the average classification accuracy of each class at the first test step.

Test	Class 0	Class 1	Class 2
2 * μ_0	$\sim 98\%$	$\sim 35\%$	$\sim 41\%$
2 * μ_1	$\sim 39\%$	$\sim 98\%$	$\sim 35\%$
2 * μ_2	$\sim 38\%$	$\sim 31\%$	$\sim 98\%$

To verify these results we show they continue regardless of optimizer and learning rate in the experiments show in the appendix, each run for 100 seeds and 100 training steps(Figure 4). To mitigate this class bias based on distance we again utilize insights gained from Neural Collapse. We apply a convex combination of the input data with the data projected to points of an ETF. Our decay parameter $\lambda = 0$ at step 0, and the previous average classification accuracy over all classes otherwise for the form:

$$\hat{x} = (\lambda x) + ((1 - \lambda) * NewPoints).$$

This appears to make convergence speed more uniform in the case of class 0. This addresses the first potential cause of bias based on the distance between data centers. But even in cases where training dynamics and convergence speed appear uniform there is still a large Max-Min margin(Figure 5). This indicates even networks with similar average convergence speed are still unfair. We identify a potential cause of this to be weight initialization. The weights could initialize unfairly every time, but this unfairness could average out between classes so they converge at the same rate, but are actually unfair during training. To test this we initialize our weights multiple times and average the initializations(Figure 6). We also set the weights to 0 manually at initialization to observe whether fairness can be achieved by starting from the same point(Figure 2).

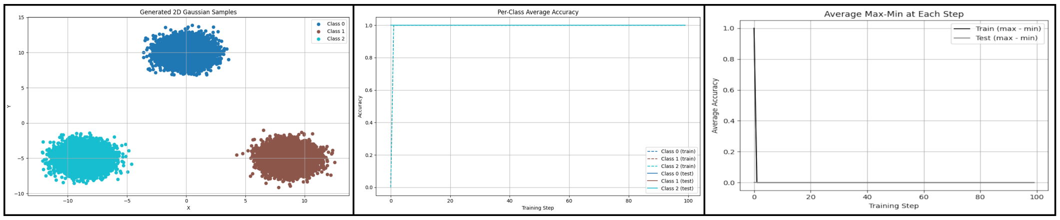

Figure 2: Initializing weights at 0

This gives us a Max-Min margin of 0 almost immediately and our classifier also converges quite quickly, showing initialization as a major cause of our observed class bias.

3.3 Results and Future Work

These tests indicate that these Max-Min biases may be explainable by the random initialization of the weights. It should also be said that the convergence speed differences observed when $\mu_i = 2 * \mu_i$ could be explainable by initialization as well. Because classes moved further from the others have additional area for their decision boundaries to be initialized in, it could weight the average convergence rate to make it appear they are being learned faster, when it fact they simply have a higher likelihood of being initialized correctly or close to correctly. Proving which factors account for these biases mathematically and how much is a promising future direction. If we understand how much each factor affects training, we can design novel regularizers to minimize bias during training. If we were able to prove initialization as the sole factor causing unfairness in our case it would serve as valuable information for practitioners who wish to mitigate bias. It would also enable further, theory based research into how muc bias in other more complicated networks is caused solely by initialization versus training dynamics. It could also indicate that methods for mitigating bias should focus on minimizing initialization disparities as well as potentially utilize projecting to an ETF for improving uniformity of average convergence.

Table 2: Summary of results

Test	Fair Convergence	Fair Max-Min
Equidistant Classes	✓	X
Far Class	X	X
Projection	✓	X
Initialization	✓	✓

4 Conclusions

Although initialization is a strong candidate as the cause for unfairness in the linear regime, experiments show it may not be the only cause. Proving how much initialization contributes to bias and whether it is the sole factor causing faster convergence may demonstrate initialization as the sole cause of unfairness in our case. Observing the tradeoff between training speed and exploring a diverse feature space has long been studied, but fairness is often left out of the conversation. There is still much work to be done in the area of optimal weight initialization to balance and understand the tradeoffs between convergence speed, exploring a diverse feature space, and enabling fair learning among all classes while taking into account regularization techniques. As such, it is high time that we begin to shift focus to the study of continuous per-class fairness.

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5 Appendix

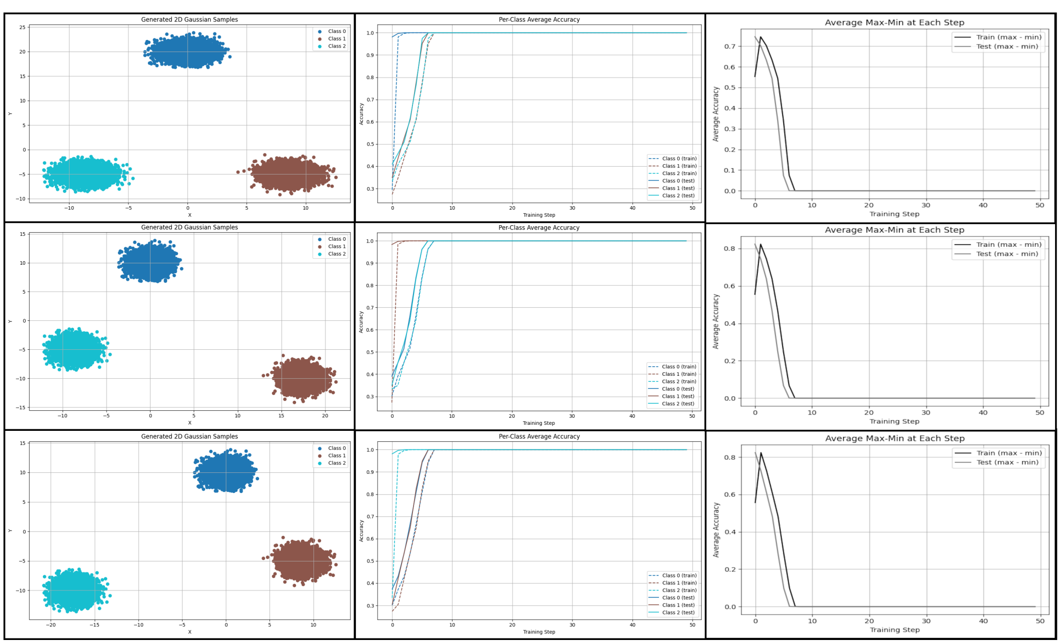

Figure 3: Examples of how moving classes can affect average convergence speed

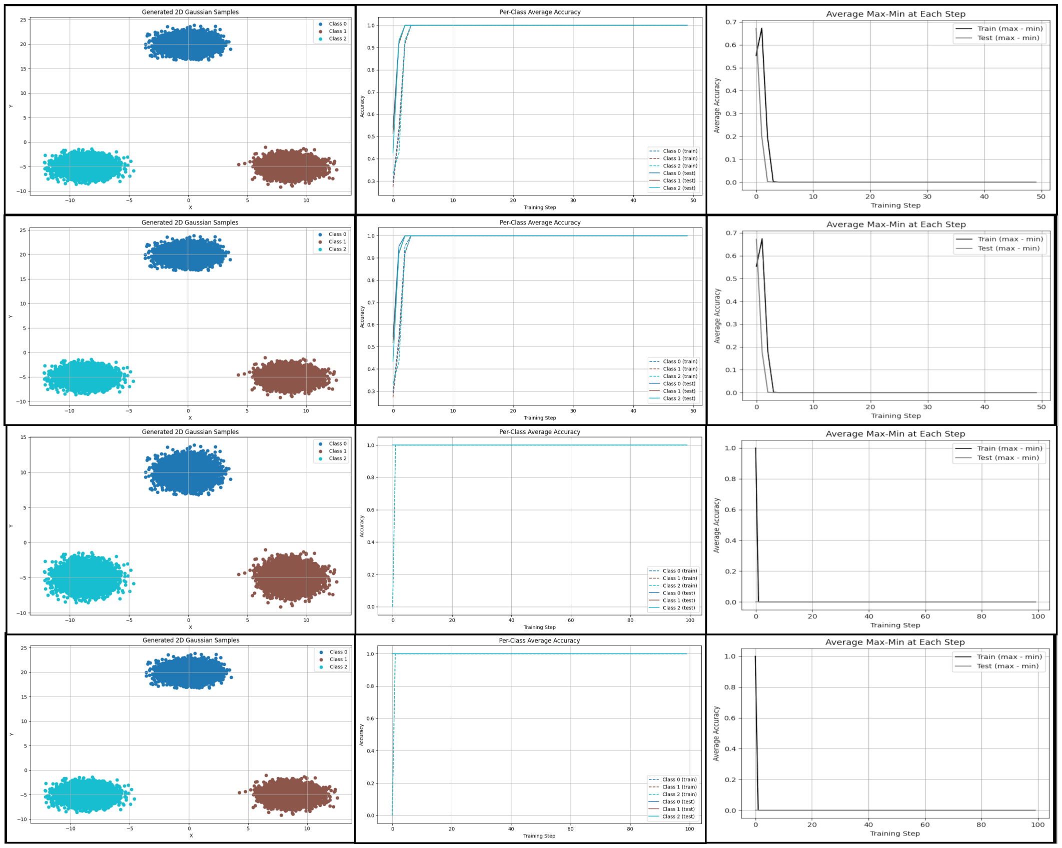

Figure 4: Top two show how projection methods can affect convergence speed and Max-Min. Bottom two show the effects of initializing weights to 0.

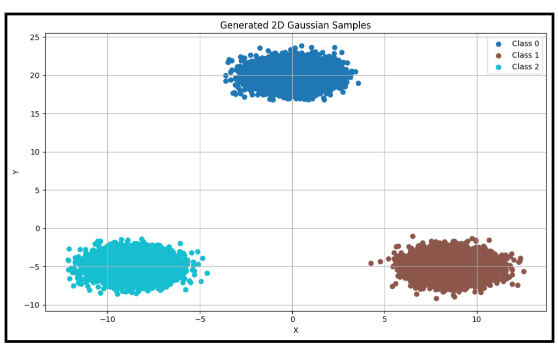

Figure 5: When varying optimizers and learning rates below, all experiments are done on this distribution of samples, with class 0 farther from the origin.

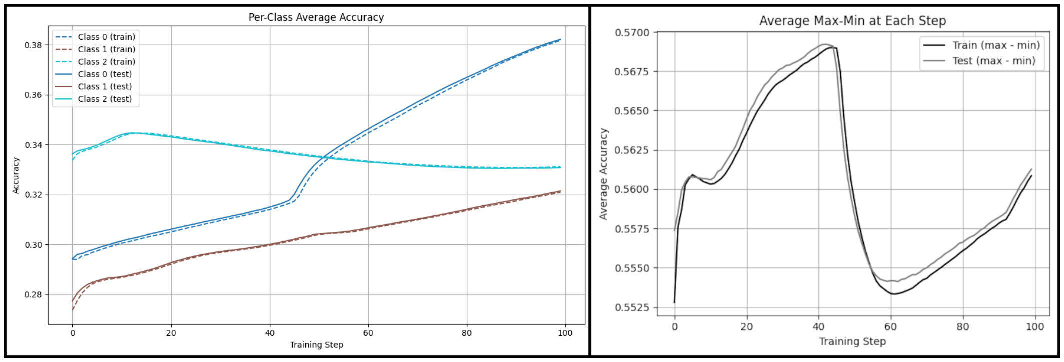

Figure 6: Adagrad optimizer with base parameters

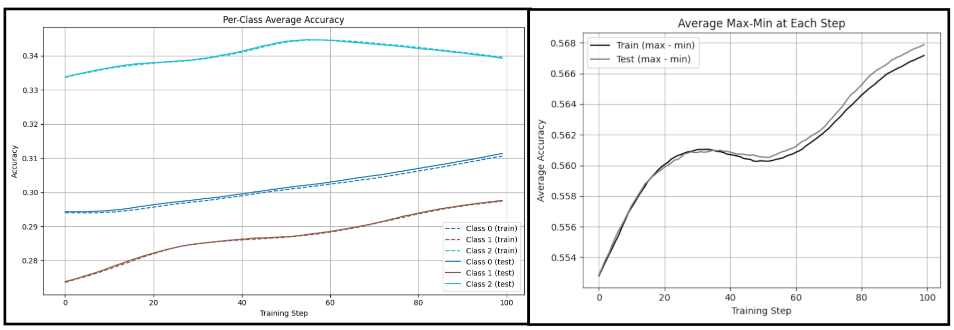

Figure 7: Adam optimizer with base parameters

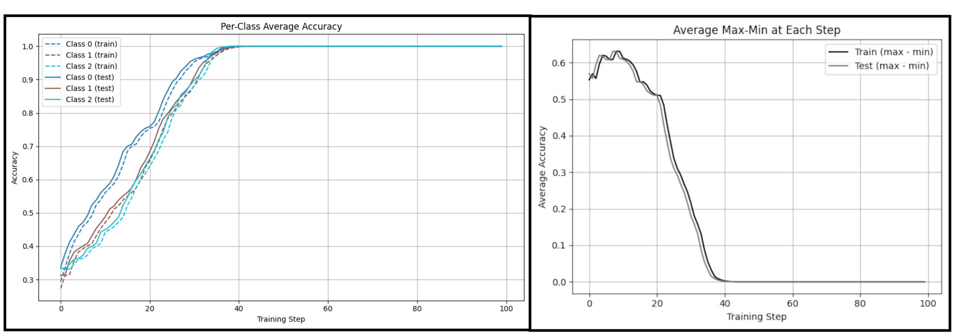

Figure 8: RMSprop optimizer with base parameters

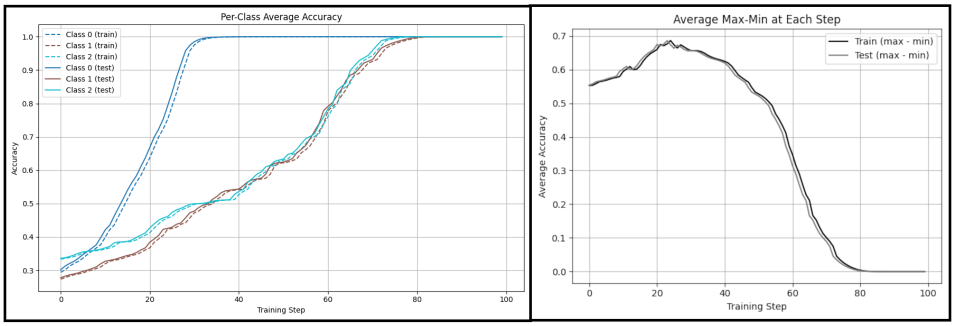

Figure 9: SGD with a base learning rate

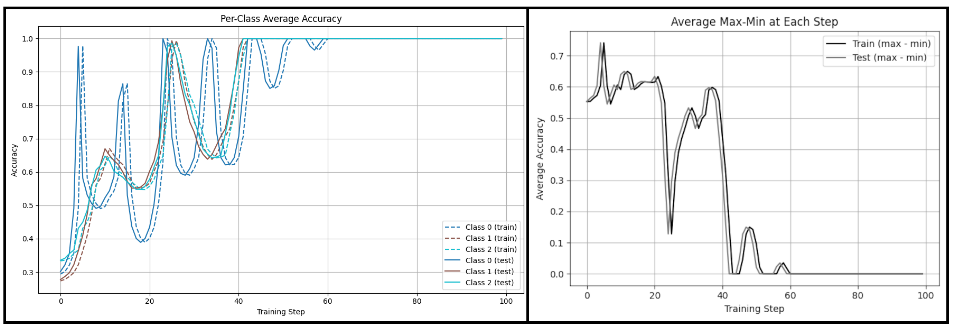

Figure 10: SGD with a base learning rate and momentum parameter of 0.9